

Lecture 01

1. Compare and contrast the roles of an Intermediate System (IS) at the Data Link, Network, and Transport layers. Provide one practical example of each type.

(4 marks)

2. Explain how the end-to-end principle and the “IP over everything” principle together form the foundation of the Internet’s design. Give a real-world example where these principles allow for easy adoption of new network technologies.

(4 marks)

3. Analyze the differences between the OSI model and the TCP/IP model in terms of layer functionality. Specifically, discuss why the Presentation and Session layers are absent in the TCP/IP model and where their functions are implemented instead.

(4 marks)

4. Describe how protocol encapsulation and multiplexing/demultiplexing work together in the TCP/IP model. Use the example of an HTTP request being sent over Ethernet to illustrate the process.

(4 marks)

5. Discuss the hierarchical structure of Internet network organization, from continental backbone networks down to individual hosts. Explain how this hierarchy supports scalability and interoperability across diverse network technologies.

Lecture 2

1. Explain how the IPv4 datagram header fields support fragmentation and reassembly across networks with different MTUs. Use a specific example involving Ethernet and X.25 networks to illustrate your explanation.

(4 marks)

2. Compare and contrast classful and classless addressing in IPv4. Discuss how CIDR alleviates the problem of address exhaustion and routing table explosion, providing an example with a given IP block.

(4 marks)

3. Describe the role of subnet masks in both subnetting and supernetting. Using the example of IP address 141.14.72.24 with subnet mask 255.255.192.0, show how the network and subnet addresses are derived and explain the impact on routing.

(4 marks)

4. Analyze the function of ICMPv4 in the IP network layer. Explain with examples how ICMP error-reporting messages (such as Destination Unreachable and Time Exceeded) assist in network diagnostics and troubleshooting.

Lecture 11

1. Compare the noise resistance and bandwidth capabilities of Twisted Pair (UTP/STP), Coaxial Cable, and Fiber Optic Cable. Explain how the physical construction of each contributes to its performance in real-world networking environments.

(4 marks)

2. Explain the principle of data transmission in Fiber Optic cables, including the roles of core, cladding, and light refraction. Differentiate between single-mode and multi-mode fiber optics in terms of application scenarios and performance.

(4 marks)

3. Analyze the use of Power Line Communication (PLC) as a transmission medium. Discuss its advantages and limitations compared to traditional wired media such as twisted pair and coaxial cable, especially in terms of deployment, data rate, and duplex mode.

(4 marks)

4. Discuss why magnetic storage media are still used for large-scale data transfer despite the availability of high-speed internet. Relate your answer to concepts such as bandwidth, cost, feasibility, and security in enterprise environments.

(4 marks)

These questions encourage students to think critically about the design, application, and trade-offs of different transmission media in modern and legacy networking contexts.

write 5 analytical question from this lecture where marks 4

Based on the content of **Lecture 11 - Transmission Media**, here are **five analytical questions** (each worth 4 marks) designed to test deeper understanding and real-world application:

1. Explain how the twisted design of UTP and STP cables reduces electromagnetic interference (EMI) and crosstalk. Compare their effectiveness and typical use cases in modern network installations.

(4 marks)

2. Describe the structural composition of a coaxial cable and explain why it supports higher bandwidth and better noise immunity compared to twisted pair cables. Provide examples of where coaxial cables are still preferred today.

(4 marks)

3. Analyze Power Line Communication (PLC) as an alternative data transmission medium. Discuss its operational principles, advantages in terms of infrastructure reuse, and limitations related to data rate and duplex communication.

(4 marks)

4. Compare single-mode and multi-mode fiber optic cables in terms of light transmission, distance capabilities, and typical applications. Explain why single-mode fiber is preferred for long-distance communication.

(4 marks)

5. Evaluate the scenarios where physical magnetic media (e.g., tapes, disks) are still used for data transfer despite the availability of high-speed networks. Consider factors such as data volume, security, cost, and network limitations.

(4 marks)

Lecture 14

1. Compare Circuit Switching and Packet Switching in terms of resource allocation, delay characteristics, and suitability for different types of applications (e.g., voice vs. data). Provide an example where each method is optimally used.

(4 marks)

2. Explain why Message Switching is no longer widely used in modern networks. Discuss its limitations in terms of storage requirements, latency, and suitability for real-time communication compared to Packet Switching.

(4 marks)

3. Describe the three phases of Circuit Switching and explain how this approach ensures reliable communication for voice applications. Contrast this with the connectionless nature of Packet Switching in the Internet.

4. Analyze how Packet Switching improves line efficiency through multiplexing and prioritization. Discuss how Quality of Service (QoS) can be implemented in a packet-switched network using priority-based forwarding.

(4 marks)

5. Compare connectionless and connection-oriented switching approaches in the context of Packet Switching. Explain the role of forwarding tables and pre-established circuits, and discuss the trade-offs in terms of setup delay and reliability.

Lecture 16

1. Compare the Stop-and-Wait and Sliding Window flow control mechanisms in terms of efficiency, resource utilization, and suitability for different network conditions. Provide a scenario where each method would be preferred.

(4 marks)

2. Explain how Automatic Repeat Request (ARQ) mechanisms handle error control in the data-link layer. Differentiate between Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ in terms of retransmission strategy and buffer requirements.

(4 marks)

3. Describe the role of positive ACK, negative ACK (NACK), and timeout in error control protocols. Explain how these mechanisms work together to ensure reliable data transmission over an unreliable medium.

(4 marks)

4. Analyze the trade-offs between Go-Back-N ARQ and Selective Repeat ARQ in terms of bandwidth efficiency, complexity, and memory usage at the receiver. In which network conditions would Selective Repeat ARQ be more advantageous?

(4 marks)

5. Illustrate with a diagram or example how the Sliding Window protocol can be combined with ARQ to achieve both flow control and error control simultaneously. Explain how window size affects performance.

Lecture 17

1. Compare and contrast Single-bit, Multiple-bit, and Burst errors in terms of their causes, detection difficulty, and impact on data integrity. Provide an example scenario where each type might occur in a communication system.

(4 marks)

2. Explain how Parity Check works for error detection, including even and odd parity schemes. Discuss its limitations and explain why it is ineffective for detecting multiple-bit or burst errors.

(4 marks)

3. Describe the Cyclic Redundancy Check (CRC) method for error detection. Explain the role of the polynomial divisor, codeword formation, and the division process at both sender and receiver. Why is CRC more robust than parity check?

(4 marks)

4. Differentiate between Backward Error Correction (BEC) and Forward Error Correction (FEC). Discuss the scenarios where each method is preferred, considering factors such as transmission medium, cost, and latency.

(4 marks)

5. Using the formula $2^r \geq m + r + 1$, calculate the minimum number of redundant bits r required for error correction of a 7-bit ASCII character. Explain the logic behind this inequality in the context of Hamming distance and error correction capability.

(4 marks)

Lecture 19

1. Explain why network addresses are considered "logical" and software-based, while MAC addresses are "physical" and hardware-based. Discuss how these two addressing layers interact in the process of Layer-2 and Layer-3 communication.

(4 marks)

2. Describe the hierarchical nature of IP addressing and its significance in routing. How does this hierarchy help routers efficiently forward packets to destinations outside a host's subnet?

(4 marks)

3. Compare and contrast Unicast, Multicast, Broadcast, and Anycast addressing modes in terms of their purpose, packet delivery mechanism, and typical use cases in modern networks.

(4 marks)

4. Explain the role of DNS in network addressing. Describe the process through which a host obtains the IP address of a remote host using DNS and how this address is used in subsequent packet forwarding.

(4 marks)

5. Analyze the routing process in a typical IP network. Discuss how routers use routing tables to determine the next hop, and why routers do not forward broadcast traffic by default.

(4 marks)

*Extra

1. Explain the process of Address Resolution Protocol (ARP) and its role in enabling communication between devices on the same broadcast domain. Discuss why ARP caching is important for network efficiency.

2. Describe the function of Internet Control Message Protocol (ICMP) in network diagnostics and error reporting. Explain how ICMP-echo and ICMP-echo-reply are used to test network connectivity and diagnose problems.

Lecture 24

1. Explain the role of TCP flags (SYN, ACK, FIN, RST, PSH, URG) in connection management, data transfer, and error handling. Illustrate how these flags are used in a typical TCP three-way handshake and connection termination.

(4 marks)

2. Describe TCP's congestion control mechanism, including Slow Start, Additive Increase/Multiplicative Decrease, and Timeout React. Explain how the window size is adjusted in response to network conditions and packet loss.

(4 marks)

3. Analyze how TCP ensures reliable data delivery through sequence numbers, acknowledgements, and retransmission timers. Discuss what happens when segments arrive out of order or are lost, and how duplicate segments are handled.

(4 marks)

4. Compare the functions of the four TCP timers: Keep-alive, Retransmission, Persist, and Timed-Wait. Explain how each timer contributes to connection stability, flow control, and crash recovery.

(4 marks)

5. Discuss how TCP multiplexing works and its impact on server performance. Explain the benefits and potential drawbacks of keeping TCP sessions open for longer periods, especially in the context of HTTP/1.1 and modern web applications.

(4 marks)

Lecture 24

1. Compare and contrast UDP and TCP in terms of reliability, connection management, header complexity, and suitability for different types of applications. Provide examples where UDP is preferred over TCP.

(4 marks)

2. Explain why UDP is considered an “unreliable” protocol and discuss the implications of this unreliability in real-time applications such as VoIP and video streaming. How does the underlying IP protocol’s “best-effort” delivery mitigate some of these issues?

(4 marks)

3. Analyze the UDP header structure and explain the purpose of each field (Source Port, Destination Port, Length, Checksum). Why is the checksum optional in IPv4, and what are the security or reliability risks of disabling it?

(4 marks)

4. Discuss the role of UDP in query-based communications and protocols like DNS and SNMP. Why is UDP’s stateless and connectionless nature advantageous in these contexts?

(4 marks)

5. Evaluate the trade-offs between using UDP and TCP for multimedia streaming. Consider factors such as latency, bandwidth efficiency, packet loss tolerance, and congestion control.

Lecture 28

1. Compare and contrast DNS and HTTP in terms of their transport layer protocols (UDP vs. TCP), communication models, and roles in the Internet infrastructure. Explain why DNS typically uses UDP despite being a critical service.

(4 marks)

2. Explain the process of email transmission using SMTP, POP3, and IMAP. Discuss the roles of User Agents, Message Transfer Agents, and mail servers, and analyze the differences between POP3's delete and keep modes.

(4 marks)

3. Describe how FTP uses out-of-band control and data connections (ports 20 and 21). Discuss the advantages and disadvantages of this approach compared to using a single connection for both control and data, as in HTTP.

(4 marks)

4. Analyze the evolution of HTTP from version 1.0 to 1.1 in terms of connection persistence. Explain the impact of persistent and non-persistent connections on web performance, server load, and latency.

(4 marks)

5. Explain why HTTP is considered a stateless protocol and discuss the mechanisms (such as cookies and sessions) used to maintain state in web applications. How does statelessness affect scalability and security?